Question				Marking details	AO1	AO2	AO3	Total	Maths	Prac
					Marks Available					
2	*(a)*	(i)		Intron {does not code / spliced out/ not translated} (1) Exon {codes/ is translated} (for a polypeptide) (1) NOT codes for an amino acid(s)	2			2		
		(ii)		Water potential falls (in the mitochondria)/ solute potential decreases (1) Water enters by <u>osmosis</u> (1) Max 1 if refer to cell		2		2		
	(b)	(i)		Phenotype parents: unaffected male, {unaffected/carrier} female (1) Genotype parents, X^DY , X^DX^d (1) Genotype gametes, X^D. Y. X^D. X^d (1) ECF Genotypes offspring, X^D Y, X^dY, X^DX^d X^DX^D (1)ECF		4		4		
		(ii)		They could {have two copies of the healthy gene/ be homozygous dominant} or be {heterozygous / carrier}/ They could be X^DX^d or X^DX^D/ can't determine until they reproduce (1)		1		1		
	(c)			Any 2 x (1) from: Cause immune response against virus/ antibodies may be produced against it (1) Problems introducing gene into muscle (cells)/ may not reach {target (cell)/ muscle (cell)}/ may invade {non target / host} cells(1) Virus may {become pathogenic/ cause disease/ cause infection/ destroy cells}(1) NOT harm/ illness May affect other genes/ reference to oncogenes(1)	2			2		
	(d)	(i)		CCGUUA(1)	1			1		

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Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
		(ii)		Shorter / different 1^{o} structure/ less amino acids/ smaller/ ORA (1)	1			1		
		(iii)		do not have to repeat treatment/ more permanent treatment/ change present in daughter cells/ change present after cell replicates/ can pass to next generation (1)	1			1		
		(iv)		Unknown long-term effects /possible activation of oncogenes / modified gene passed on to next generation/ affect {other genes/ later generations}(1)	1			1		
	(e)			Not sex linked / on autosomes (1) Dominant/not recessive (1) Three examples to justify the conclusions for (3) If recessive 1 + 2 could not produce 6/ unaffected {female/ child} 8 + 9 could not produce 13 /14/ unaffected {female/ child} OR must be dominant, otherwise all children of 1+ 2 / 8 + 9 would be affected If sex linked 1 + 2 could not produce 6/ unaffected female 8 + 9 could not produce 13 or 14/ unaffected female 4 + 5 could not produce 10/ an affected male OR must be autosomal, otherwise all **daughters** of 1+ 2 / 8 + 9 would be affected Max (4) if examples are given to only support one conclusion.			5	5		
				Question 2 total	**8**	**7**	**5**	**20**		

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